WSABI Summer Research Lab **Summer 2026**

Lab 15: Regularization and Ridge Regression

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15.1 NBA Lineup Data and Player Effects

Use `15_nba-lineups.rds` to estimate player impact from 2023–24 NBA possession data. Each row is one possession. The variable `pts_poss` is the number of points scored by the team with the ball, `lineup_team` gives the five offensive players, and `lineup_opp` gives the five defensive players.

The starter code handles the data preparation: parsing lineups, keeping complete possessions, creating player indices, building sparse design matrices, and creating chronological training, validation, and test splits. Your work should focus on fitting and comparing the plus-minus models.

This lab has two versions of the plus-minus design. The first gives player j one total-impact coefficient β_j. For possession i, define

$$x_{ij}^{\text{total}} = \begin{cases} 1, & \text{player } j \text{ is on offense on possession } i, \\ -1, & \text{player } j \text{ is on defense on possession } i, \\ 0, & \text{otherwise.} \end{cases}$$

The model is

$$y_i = \alpha_0 + \sum_{j=1}^{m} x_{ij}^{\text{total}} \beta_j + \epsilon_i.$$

Positive coefficients mean that, after accounting for the other players on the floor, the player's team tends to score more when he is on offense and allow fewer points when he is on defense.

The second version gives each player separate offensive and defensive coefficients:

$$\beta_j^{\text{off}} \qquad \text{and} \qquad \beta_j^{\text{def}}.$$

Use the lecture sign convention:

$$y_i = \alpha_0 + \sum_{j \in OP(i)} \beta_j^{\text{off}} - \sum_{j \in DP(i)} \beta_j^{\text{def}} + \epsilon_i.$$

Larger offensive coefficients mean stronger scoring impact. Larger defensive coefficients mean stronger point prevention because they subtract from the offense's point total.

15.2 Regularization and Model Selection

Fitting the plus-minus model by ordinary least squares gives adjusted plus-minus (APM). Basketball lineup data are highly collinear, so APM can give extreme estimates to players who mostly share possessions with the same teammates.

Regularized adjusted plus-minus (RAPM) uses ridge regression:

$$\widehat{\boldsymbol{\beta}}^{\text{ridge}} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_i (y_i - x_i^T \boldsymbol{\beta})^2 + \lambda \sum_j \beta_j^2 \right\}.$$

The tuning parameter λ controls how strongly player effects are pulled toward zero.

Choose λ using the prepared held-out split:

1. Fit candidate ridge models on the training games over a grid of λ values.
2. Choose λ using validation RMSE.
3. Evaluate on the test games only after the modeling procedure is finalized.

Do not use the test set while choosing λ, selecting players for plots, or deciding how to change the model.

15.2.1 Task 1: Fit Combined Plus-Minus Models

Using the prepared design matrix `X_combined`, fit:

1. a mean-only baseline;
2. raw plus-minus for each player;
3. unregularized APM;
4. ridge RAPM over a grid of candidate λ values.

Choose the combined RAPM λ using validation RMSE. Report the selected λ and the validation RMSE for the baseline, APM, and RAPM.

15.2.2 Task 2: Visualize Shrinkage

Choose at least ten players for the visual. Plot raw plus-minus, APM, and RAPM estimates from the combined model. Report estimates in points per 100 possessions and include a zero line. Sort the plot by RAPM.

Explain which estimates move most under ridge regularization. Are the largest APM estimates supported by many possessions, or do they look like noisy credit assignment?

15.2.3 Task 3: Fit an Offense-Defense RAPM Model

Using the prepared split design matrix `X_split`, fit ridge models over the same λ grid and choose λ using validation RMSE. Make a plot for your selected players showing offensive and defensive estimates separately in points per 100 possessions. Identify at least one player whose total-impact estimate hides a meaningful offense-defense difference.

15.2.4 Task 4: Final Test Evaluation

After all choices above are fixed, evaluate the final models on the test games. Report test RMSE for the mean-only baseline, combined APM, combined RAPM, and split RAPM.

Which model predicts held-out possessions best? Explain what the split model reveals that the combined model hides, and whether the extra flexibility appears useful for prediction.